

MasterTweak — Quick Start

MasterTweak Team

MasterTweak Quick Start Guide (English)

Welcome to MasterTweak! This guide walks you through the complete workflow: open an audio file, choose a preset, and export a mastered version in just 7 simple steps.

Getting Started in 7 Steps

Step 1: Launch MasterTweak

Double-click the MasterTweak application icon on your desktop or in your Applications folder.

On **Linux**, run from the terminal:

```
./MasterTweak
```

The main window appears with an empty project. You will see: - **File browser** panel (left) - **Preset selector** (top-left dropdown) - **Parameter override panel** (right) with sliders for EQ, compression, saturation, etc. - **Waveform display** (center, greyed out until you load audio) - **Render button** (bottom-right, disabled until audio is loaded)

Step 2: Open an Audio File

1. Click **Open File** or use **File → Open**.
2. Navigate to your WAV, FLAC, or AIFF file.
3. Click **Open**.

The file is loaded and analyzed automatically. You will see: - Waveform displayed in the center panel - **7-band frequency analysis** in the background (red spikes show measured RMS per band) - **Analysis results** displayed below the waveform (RMS per band, loudness, dynamics, etc.) - **Render button** becomes **enabled** (bright blue)

Step 3: Choose a Preset

A preset is a recipe for mastering—it defines target loudness, EQ targets, compression shape, and dynamic character.

In the **Preset dropdown** (top-left), select one of the bundled presets:

Preset	Use Case
Hip-Hop / Rap	Aggressive, punchy. Tight low mids, loud low end, strong limiting.
Rock	Balanced, transient-heavy. Clear highs, controlled lows.
Electronic / EDM	Bright, wide stereo. High-passed subs, airy top end.
Pop	Commercial, smooth. Warm mids, compressed, modern loudness.
Jazz / Acoustic	Open, dynamic. Minimal EQ, gentle compression.
Classical	Natural, reference. Minimal saturation, linear response.
Podcast / Voice	Clear, intelligible. Boost presence band (2–6 kHz), multiband comp on sibilants.

Tip: Start with a preset that matches your genre. The analysis automatically feeds into the DSP chain, so you don’t need to tweak anything—but you can if you want to.

Step 4: Review Parameters (No Changes Required)

You are not required to change anything. Each parameter shows: - **Current value** (slider position) - **Default recommendation** (calculated from your audio + preset) - **Range** (e.g., –12 to +12 dB)

If you want to listen to how the default advice sounds: - Click **Preview** (or press Space) - The audio renders in real-time and plays through your speakers - Listen to the A/B comparison (see Step 6)

To customize (optional): - Drag any slider to override the recommendation - The preview re-renders when you release the slider - You can always click **Reset to Defaults** to go back

Step 5: Click Render

Once you are satisfied with the preview (or ready to accept the defaults), click the large **Render** button (bottom-right).

A progress dialog appears: - “Rendering... 50% complete” - Two-pass rendering: analyze + apply DSP chain - This may take 10–30 seconds depending on file length and CPU

When done: - **Render button** changes to “**Done! Ready to save**” (green) - **Export button** becomes **enabled** (top-right)

Step 6: Listen and Compare (A/B)

Before saving, you can A/B the before and after:

- **Waveform view** shows the **original (blue) vs. mastered (red)** side-by-side
- Click **A/B Toggle** (or press **T**) to switch between:
 - Playing the **original** file
 - Playing the **mastered** file (from the in-memory render buffer)
 - Each plays the first 10 seconds on repeat so you can listen closely

What to listen for: - Bass is tighter, more present (but not muddy) - Kick/snare/vocals sit well in the mix - High end is bright but not harsh - Overall loudness is consistent across tracks - Dynamics are controlled without sounding compressed - Stereo image is open and balanced

If you want to tweak and re-render, just adjust sliders and click **Render** again.

Step 7: Save

When you are happy with the result:

1. Click **Export** (or **File → Export**)
2. Choose output settings (see below)
3. Pick a filename and location
4. Click **Save**

The mastered file is written to disk.

Common Options

Output Format

At the bottom of the export dialog, you will see:

- **Bit Depth:**
 - **16-bit** (CD quality, smallest file, used for distribution)
 - **24-bit** (studio quality, larger file, used for archiving)
 - **32-bit float** (mixing quality, rarely needed for final output)
- **Format:**
 - **WAV** (universal, uncompressed)
 - **FLAC** (lossless, 30–50% compression, recommended for archiving)

Target Loudness Platform

Select the platform your mastered track will be delivered to. This controls the limiter ceiling:

Platform	Target (LUFS)	Notes
Spotify	−14 LUFS	Normalized by algorithm; overkill to exceed this.
Apple Music	−16 LUFS	Legacy loudness standard.
YouTube	−13 LUFS	Video platform; slightly louder than Spotify.
Streaming (Generic)	−14 LUFS	Safe middle ground for all streamers.
Broadcast	−23 LUFS	Radio, TV, podcasts. Much quieter.
CD / Reference	−9 LUFS	Uncompressed, dynamic, for archival or audiophile playback.

How it works: The limiter’s ceiling moves to match your target platform. A higher target (e.g., −9 LUFS for CD) allows louder peaks; a lower target (e.g., −23 LUFS for broadcast) squashes dynamics more.

Export Advice Button

Click **Export Advice** to save a detailed Markdown report of what MasterTweak did: - Measured RMS, correlation, and crest factor per band - EQ gains and Q values applied - Compression ratios and thresholds - Saturator drive - Limiter settings - Warnings (e.g., “High crest factor; consider more compression”)

This is useful for: - Learning what the DSP did - Replicating settings in your DAW - Troubleshooting (if something doesn’t sound right)

MIDI Control (Optional)

If you have a MIDI controller (keyboard, mixer, knobs), you can map physical faders to MasterTweak parameters.

Setup

1. Open **Settings** → **MIDI**.
2. Select your MIDI device from the list (e.g., “Akai APC40”, “Behringer FCB1010”).
3. Click **Learn** next to each parameter.
4. Move the corresponding slider/knob on your hardware.
5. MasterTweak records the mapping.
6. Click **Save**.

Mappings

Once set up, you can control in real-time while previewing:

Parameter	Range	Use
EQ Gain (per band)	−12 to +12 dB	Tweak low-end punch, presence, air
Multiband Comp Threshold (per band)	−30 to 0 dB	Control when compression kicks in
Saturator Drive	0 to 6 dB	Add harmonic character (tape, tube simulation)
Limiter Ceiling	−1 to −6 dBTP	Emergency loudness limit
Stereo Width	0 to 200%	Narrow (mono) to wide (exaggerated)

Frequency Band Reference

MasterTweak uses a 7-band split with fixed crossovers. Learn what each band does:

Band	Range	Crossovers	Purpose
Sub	20–80 Hz	—	Deep bass, kick fundamentals, sub-bass rumble.
Lows	80–250 Hz	80 Hz	Kick body, bass body, warmth.

Band	Range	Crossovers	Purpose
LowMids	250–500 Hz	250 Hz	Mud zone, bass clarity, guitar/vocal body.
Mids	500 Hz–2 kHz	500 Hz	Vocal presence, instrument clarity.
HiMids	2–6 kHz	2 kHz	Presence, sibilance, instrument edge.
Highs	6–16 kHz	6 kHz	Brightness, hi-hat, cymbals, air.
Air	16 kHz+	16 kHz	Ultra-highs, sparkle, aliasing zone.

When to boost/cut: - Boost a band if it measures too quiet (low RMS) vs. the preset target - Cut a band if it measures too loud or harsh - Use narrow bell filters (not shelves) for surgical control - Sub and Air often work better as shelves (broad, smooth)

Advanced: Bypass Stages

In the CLI or via environment variables, you can disable entire DSP stages for troubleshooting or creative use:

```
./mastertweak --bypass-eq input.wav output.wav
./mastertweak --bypass-comp input.wav output.wav
./mastertweak --bypass-sat input.wav output.wav
./mastertweak --bypass-width input.wav output.wav
./mastertweak --bypass-limiter input.wav output.wav
```

Use cases: - `--bypass-eq`: Hear unEQ'd response; debug if preset EQ is wrong. - `--bypass-comp`: Hear uncompressed dynamics. - `--bypass-sat`: Remove harmonic character (make it clean/sterile). - `--bypass-width`: Collapse to mono (check mono compatibility). - `--bypass-limiter`: Let audio clip to see raw DSP output (risky).

CLI Mode: Batch Processing

For batch mastering or unattended renders, use the CLI:

```
mastertweak --input song1.wav --output song1-mastered.wav --preset "Pop"
mastertweak --input song2.flac --output song2-mastered.flac --preset "Hip-Hop / Rap"
```

Common flags: - `--preset <name>` — Use a bundled preset - `--output-format wav` or `flac` — Default is WAV - `--bit-depth 16` or `24` — Default is 16 - `--target-platform spotify` — Loudness ceiling; default is “Streaming (Generic)” - `--analyze-only` — Print analysis, skip render (useful for debugging) - `--override-eq <band> <gain>` — Override EQ gain on a single band (0-6) - `--override-sat-drive <db>` — Override saturator drive

Example batch script:

```
#!/bin/bash
for file in *.wav; do
    mastertweak --input "$file" --output "mastered/$file" --preset "Pop"
done
```

Quick Troubleshooting

Problem	Cause	Solution
File won't open	Unsupported format or corrupted file.	Confirm file is WAV, FLAC, or AIFF. Try converting with FFmpeg: <code>ffmpeg -i input.mp3 output.wav</code>
No waveform appears	File decoded but playback codec missing.	Check system audio settings; ensure PulseAudio or ALSA is working.
Render is very slow	Large file (>30 min) or slow CPU.	Expected for large files. Close other apps. Release version is faster than Debug.
Output sounds distorted	Limiter ceiling is too low for the content.	Raise target platform or manually increase limiter ceiling slider.
Output is too quiet	Loudness platform is too conservative.	Lower target platform (e.g., from “Broadcast” to “Streaming”).
Crash on render	Out of memory or corrupted file.	Free up RAM. Restart app. Try a smaller test file.
A/B playback skips	Audio device buffer underrun.	Increase buffer size in Settings → Audio.

Next Steps

This guide covers the essentials. For deeper dives:

- **Full User Manual** ([docs/user/manual_en.md](#)) — Detailed parameter explanations, DSP algorithm reference, preset-building workflow
- **Preset Builder Guide** ([docs/user/preset_builder_en.md](#)) — Create custom presets from scratch using MasterTweak's Python tools
- **CLI Reference** ([docs/user/cli_reference_en.md](#)) — All command-line flags and scripting examples
- **FAQ** ([docs/user/faq_en.md](#)) — Common questions and answers

Happy mastering!

MasterTweak Version 1.0 — Standalone offline auto-mastering, driven by MixAdvice presets.

For bug reports or feature requests, visit the project GitHub issues page.